Notes: Sias (RFS, 2004)
Institutional Herding

Contents

1 Big picture

Sias (2004) asks whether institutional investors herd, meaning whether they follow each other into and out of the same securities over time. The paper studies quarterly 13F holdings from March 1983 to December 1997 and measures whether institutional demand for a stock in one quarter predicts institutional demand for the same stock in the next quarter.

The key finding is that institutional demand is strongly persistent. The fraction of institutions buying a stock this quarter is positively correlated with the fraction buying the same stock last quarter. This persistence has two sources: institutions follow their own lag trades, and institutions follow other institutions' lag trades. The second channel is the paper's direct measure of institutional herding.

The paper's interpretation is mainly information-based. Momentum trading exists but explains little of the herding. Institutional demand is weakly positively related to future returns rather than negatively related to future returns, so herding does not appear to push prices away from fundamentals and then reverse. Herding is strongest in small-capitalization stocks, which is most consistent with informational cascades: institutions infer information from each other's trades, especially when private signals are noisier.

2 Incremental contribution of the paper

1. **Introduces a direct interquarter test of institutional herding.** Earlier work largely used the Lakonishok, Shleifer, and Vishny measure, which detects whether institutions are unusually concentrated on one side of a trade within a period. Sias instead tests whether institutions follow one another across adjacent quarters.
2. **Separates herding from own-trade persistence.** The cross-sectional correlation between institutional demand this quarter and last quarter is decomposed into two parts: individual institutions following their own previous trades and institutions following other institutions' previous trades.
3. **Shows that herding is large even when conventional measures look small.** The paper finds LSV-style herding measures similar to previous studies, but its direct correlation-based approach reveals substantial herding.
4. **Distinguishes herding from habit investing and momentum trading.** By studying return-adjusted portfolio weights and controlling for lag returns, the paper shows that herding is not primarily driven by fund flows or by institutions simply chasing winners.
5. **Connects herding to price informativeness.** The paper tests whether herding is followed by return reversals. It finds no such reversal, which is more consistent with information-based herding than with fads, reputational herding, or characteristic herding that drives prices away from fundamentals.
6. **Documents heterogeneity by firm size, time, and investor type.** Herding exists across capitalization groups and investor classes, but it is strongest in small firms and varies across bank trust departments, insurance companies, mutual funds, independent advisors, and unclassified institutions.

3 Research question and conceptual framework

3.1 What is herding?

The paper defines herding as a group of investors following each other into or out of the same securities over a period of time. Because trades occur sequentially, the key object is temporal dependence in the cross section of trades: do institutions buy stocks this quarter that other institutions bought last quarter?

This definition differs from simply observing many buyers in the same stock at the same time. A stock may have many institutional buyers by chance. The paper therefore focuses on whether institutional demand is predictable from lag institutional demand.

3.2 Motives for institutional herding

The paper discusses five theoretical explanations.

- **Informational cascades:** institutions infer information from other institutions' trades and follow them.
- **Investigative herding:** institutions receive correlated signals because they analyze similar information or follow similar research processes.
- **Reputational herding:** managers avoid deviating from the crowd because being wrong alone is reputationally costly.
- **Fads:** institutions collectively move into or out of securities for nonfundamental reasons.
- **Characteristic herding:** institutions prefer similar characteristics, such as momentum stocks, liquidity, size, or prudence-compatible holdings.

The paper's tests cannot perfectly identify one mechanism, but the combination of no return reversals, stronger herding in smaller stocks, and limited explanatory power of momentum points most strongly toward informational cascades.

4 Data

The study combines CRSP monthly data with CDA-Spectrum institutional holdings data from 13F filings. The sample runs from March 1983 through December 1997, giving 60 quarters of holdings and 59 quarters in which demand changes can be measured.

Institutional investors with at least $100 million under management in exchange-traded or NASDAQ equity securities must file 13F reports. The holdings are aggregated across units within an institution. For example, Fidelity Management files one 13F report covering all Fidelity funds. This aggregation means the paper observes institution-level trading, not individual fund-level trading.

CDA-Spectrum classifies institutions into five groups:

- bank trust departments,
- insurance companies,

- mutual funds,
- independent investment advisors,
- unclassified institutions.

An institution is a buyer of stock k in quarter t if its ownership fraction in the stock increases over the quarter. It is a seller if its ownership fraction decreases. If the position or share ownership fraction is unchanged, the institution is not counted as a trader in that stock-quarter.

5 Measurement and empirical design

5.1 Raw fraction of institutions buying

For stock k in quarter t, the raw fraction of institutions buying is

$$RawA_{k,t} = \frac{\#\text{ institutions buying } k \text{ in } t}{\#\text{ institutions buying } k \text{ in } t + \#\text{ institutions selling } k \text{ in } t}.$$

The variable is then standardized within each quarter:

$$A_{k,t} = \frac{RawA_{k,t} - \overline{RawA_t}}{\sigma(RawA_{k,t})}.$$

Standardization puts all quarterly cross sections on the same scale. Because both dependent and independent variables have zero mean and unit variance, regression slopes can be read as cross-sectional correlations.

5.2 Baseline herding regression

The central regression is estimated separately each quarter across stocks:

$$A_{k,t} = \beta_t A_{k,t-1} + \varepsilon_{k,t}.$$

If institutions neither follow their own trades nor follow other institutions' trades, the average β_t should be zero. A positive average β_t means institutional demand is persistent across quarters.

5.3 Decomposition of persistence

The slope β_t can be decomposed into:

1. a component from institutions following their own lag trades, and
2. a component from institutions following other institutions' lag trades.

The first component is positive when the same institution tends to buy the same stock in consecutive quarters or sell the same stock in consecutive quarters. The second component is positive when institutions buy stocks that other institutions bought last quarter or sell stocks that other institutions sold last quarter. This second component is the direct measure of herding.

5.4 Return-adjusted portfolio weights

A possible alternative explanation is habit investing: institutions may receive persistent fund flows and mechanically invest those flows in their existing portfolios. To test this, the paper studies whether institutions increase return-adjusted portfolio weights. The return-adjusted benchmark asks what the portfolio weight would have been if the institution did not rebalance during the quarter.

An institution is treated as increasing the return-adjusted portfolio weight in stock k if

$$\frac{V_{n,k,t}}{\sum_k V_{n,k,t}} > \frac{V_{n,k,t-1}(1+R_{k,t})}{\sum_k V_{n,k,t-1}(1+R_{k,t})}.$$

If herding disappears using return-adjusted portfolio weights, then the baseline results could be driven by correlated fund flows. If it remains, then institutions are actively tilting toward the same stocks over time.

6 Main results

6.1 Baseline persistence and herding

The baseline correlation between institutional demand this quarter and last quarter is 0.1194 for stocks with at least one institutional trader. Roughly half is due to institutions following their own previous trades and half is due to institutions following other institutions' previous trades. Among securities with at least five institutional traders, the total correlation rises to 0.1755, and the herding component rises to 0.1081.

This is the paper's core result: institutional investors both continue their own trading programs and follow each other into and out of securities.

6.2 Comparison with the Lakonishok-Shleifer-Vishny measure

The paper computes the Lakonishok, Shleifer, and Vishny measure in the same sample and obtains an average of 0.0178 for all stock-quarters with at least one institutional trader. This is close to values in earlier studies, so the difference between Sias's findings and previous conclusions is not mainly due to sample choice. It comes from methodology. The LSV measure indirectly infers herding within a period, while Sias directly measures whether investors follow one another across quarters.

6.3 Habit investing is not the main explanation

Using return-adjusted portfolio weights, the paper again finds strong persistence. For stocks with at least one institutional trader, the total correlation is 0.1195, almost identical to the baseline. For stocks with at least 20 institutional traders, the correlation rises to 0.2959, with a large herding component of 0.2513.

These results are inconsistent with the view that herding is mainly caused by correlated flows and proportional rebalancing into existing portfolios.

6.4 Momentum trading explains little of herding

Institutions are momentum traders: lag returns positively predict institutional demand. However, lag institutional demand remains much more important. For all stocks with at least one institutional trader, the coefficient on lag institutional demand is 0.1133 after controlling for lag returns, compared with 0.0691 for lag returns. Thus institutional demand is more strongly related to its own lag than to lag returns.

Momentum trading therefore exists, but it accounts for little of institutional herding.

6.5 Institutional demand and subsequent returns

Institutional demand is positively related to previous-quarter and same-quarter returns. More important for interpretation, it is not negatively related to future returns. For stocks with at least five, ten, or twenty institutional traders, institutional demand is positively related to returns over the next quarter, six months, and year.

If herding were driven mainly by fads, reputational pressure, or nonfundamental characteristic demand, one might expect return reversals. The absence of reversals supports the interpretation that herding is connected to information being incorporated into prices.

6.6 Herding by firm size

Herding appears in all capitalization quintiles, but average herding contributions are strongest in small-capitalization stocks. The average herding contribution declines monotonically from 0.0036 in the smallest quintile to 0.0001 in the largest quintile.

This pattern helps distinguish informational cascades from investigative herding. If herding came mainly from correlated research signals, it might be stronger in large stocks where information is more precise and widely processed. Instead, herding is strongest in small stocks, where private signals are noisier and institutions may rationally place more weight on other institutions' trades.

The paper also finds that institutions follow their own trades more in smaller stocks, consistent with trading-cost explanations. Building or liquidating positions gradually may be more important in less liquid securities.

6.7 Changes over time

The paper divides the sample into two periods: September 1983-September 1990 and December 1990-December 1997. The overall correlation between institutional demand and lag institutional demand declines from 0.1982 to 0.1527 for stocks with at least five institutional traders. The decline is mainly due to a fall in the own-trade component from 0.0823 to 0.0524.

Average contribution measures also show a decline over time. The paper interprets the decline in own-trade persistence as consistent with improved market liquidity reducing the need to accumulate or liquidate positions slowly. The decline in herding is concentrated especially in large stocks, possibly because information became incorporated more quickly in those stocks.

6.8 Herding by investor type

All institutional investor types exhibit statistically significant herding and own-trade persistence. The magnitudes differ substantially. Bank trust departments and independent advisors show large total correlations. In average contribution terms, bank trust departments have the highest herding contribution, while mutual funds and insurance companies have the weakest herding contributions.

The evidence for reputational or characteristic herding by investor class is mixed. Bank trust departments and insurance companies are more likely to follow institutions of the same type, but mutual funds and independent advisors do not show a strong tendency to follow their closest competitors. Since mutual funds and independent advisors should be especially sensitive to reputation, this weakens a pure reputational herding interpretation.

7 Interpretation

The paper's preferred interpretation is that institutions herd because they infer information from each other's trades. Three pieces of evidence are especially important.

First, the herding component remains large after controlling for lag returns, so momentum trading is not the main source of herding. Second, institutional demand is not followed by return reversals, so herding does not appear to be merely nonfundamental crowding. Third, herding is strongest in small-capitalization stocks, where private information is noisier and institutions are more likely to learn from the actions of others.

The own-trade component has a different interpretation. Institutions appear to accumulate or liquidate positions gradually, especially in smaller stocks. This is consistent with trading costs and market impact: executing a large order all at once may reveal information or require a liquidity premium, so institutions spread trades over time.

8 What the paper does not show

The paper is careful not to claim that all herding is irrational. The results are more consistent with rational information aggregation than with fads. It also does not observe intraperiod trade sequencing or individual fund-level trades inside a 13F filer, so some finer mechanisms cannot be tested directly. For example, mutual fund herding inside a fund family is not observable because 13F filings aggregate holdings at the institution level.

The paper also studies quarterly holdings rather than transaction-level data. A manager can buy and sell within a quarter without changing quarter-end holdings, so the measures capture persistent changes in institutional positions rather than all trading activity.

9 Extracted tables from the paper

9.1 Table 1: Descriptive Statistics

Read. Table 1 describes the 13F universe over time. The number of institutions grows from 556 in June 1983 to 1,274 in December 1997. Independent investment advisors drive much of this growth, rising from 173 to 892. The average institution holds 273 securities, and the number of securities with institutional trading grows steadily over the sample.

Interpretation. The institutional investor universe becomes much larger and more advisor-dominated over time. This motivates later tests of whether herding changes over time and differs by investor type.

9.2 Table 2: Tests for Herding - Buyer if Increased Position

Read. Table 2 reports the baseline regression of institutional demand on lag institutional demand. For all securities with at least one institutional trader, the average coefficient is 0.1194; the own-trade component is 0.0617 and the herding component is 0.0576. For securities with at least five institutional traders, the coefficient rises to 0.1755, with a larger herding component of 0.1081.

Interpretation. Institutional demand is persistent. About half of the baseline persistence is true herding, and the evidence strengthens when the sample is restricted to stocks with more institutional trading.

Table 1
Descriptive statistics

	Average all periods	June 1983	Dec. 1986	Sept. 1990	June 1994	Dec. 1997
Panel A: Number of institutional investors						
No. of Institutions	894	556	703	908	1074	1274
No. of Banks	189	198	190	200	189	164
No. of insurance companies	66	61	63	68	70	68
No. of mutual funds	58	46	50	56	61	81
No. of independent advisors	502	173	325	491	681	892
No. of Unclassified	79	78	75	93	73	69
Panel B: Average number of securities with:						
≥1 trader	5505	3863	4697	5040	6512	7437
≥5 traders	3783	2246	3055	3262	4710	5745
≥10 traders	2851	1654	2209	2432	3658	4464
≥20 traders	1966	1143	1430	1637	2519	3308
≥5 bank traders	4623	1563	1979	1812	2867	3267
≥5 insurance company traders	3080	500	569	787	1261	1898
≥5 mutual fund traders	2863	366	323	641	1150	2481
≥5 independent advisor traders	5033	1522	2215	2616	3795	4742
≥5 unclassified traders	2305	320	534	667	882	1029
Panel C: Average number of securities held by each manager						
All institutions	273	224	249	259	300	332
Banks	440	293	391	395	521	625
Insurance companies	329	202	211	295	422	567
Mutual funds	313	182	174	241	368	659
Independent advisors	211	213	201	203	220	227
Unclassified	245	112	183	245	294	383

Each quarter between March 1983 and December 1997 we calculate the number of institutional investors (overall and by type), the number of securities traded by at least 1, 5, 10, or 20 institutional investors, the number of securities traded by at least 5 institutional investors of each type of institution, and the cross-sectional average number of securities held by each institutional investor (overall and by manager type). Panels A, B, and C report the time-series averages of these figures as well as their values (at approximately) every 15 quarters. Because 13F reporting is aggregated across different units within an institution, the number of institutions reflects the number of unrelated institutions buying or selling the security. Fidelity Management, for example, files a single 13F report that covers all Fidelity funds.

Table 1: Descriptive Statistics

Table 2
Tests for herding—buyer if increased position

$$\Delta_{k,t} = \beta_t \Delta_{k,t-1} + \varepsilon_{k,t}$$

	Partitioned slope coefficient		
Average coefficient (β)	Institutions following their own trades	Institutions following others' trades	Average R^2
Panel A: Securities with ≥1 institutional trader			
0.1194 (12.67)**	0.0617 (7.33)**	0.0576 (10.12)**	1.93%
Panel B: Securities with ≥5 institutional traders			
0.1755 (25.54)**	0.0674 (17.72)**	0.1081 (18.90)**	3.05%
Panel C: Securities with ≥10 institutional traders			
0.1727 (30.78)**	0.0554 (23.32)**	0.1173 (23.62)**	3.16%
Panel D: Securities with ≥20 institutional traders			
0.1602 (29.86)**	0.0414 (25.42)**	0.1188 (23.96)**	2.73%

For each security and quarter between March 1983 and December 1997 we calculate the fraction of institutional traders that increase their position in the security. An investor is defined as increasing their position if they hold a greater fraction of the firm's shares at the end of the quarter than they held at the beginning. All data are standardized (i.e., rescaled to zero mean, unit variance) each quarter. We then estimate 58 quarterly cross-sectional regressions of institutional demand on lag institutional demand. Because there is a single independent variable in each regression and the data are standardized, these regression coefficients are also the cross-sectional correlations between institutional demand and lag institutional demand. The first column reports the time-series average of these 58 correlation coefficients and associated t-statistic (in parentheses, computed from time-series standard errors). The second and third columns report the portion of the correlation [see Equation (4)] that results from institutional investors following their own lag trades and the portion that results from institutions following the previous trades of other institutions (herding). Panels B, C, and D report the averages when limiting the sample to securities with at least 5, 10, or 20 institutional traders, respectively. **Indicates statistical significance at the 1% level; *indicates statistical significance at the 5% level.

Table 2: Tests for Herding - Buyer if Increased Position

9.3 Table 3: Tests for Herding - Buyer if Increased Return-Adjusted Portfolio Weight

Read. Table 3 repeats the herding tests using return-adjusted portfolio weights. The coefficient is 0.1195 for all securities with at least one institutional trader and rises monotonically with the minimum number of traders, reaching 0.2959 for securities with more than 20 institutional traders.

Interpretation. The persistence and herding results are not mechanically generated by fund flows into institutions and proportional reinvestment in existing portfolios. Institutions actively increase the relative portfolio weight of stocks that other institutions previously increased.

9.4 Table 4: Standardized Regression of Institutional Demand on Lag Institutional Demand and Lag Return

Read. Table 4 adds lag return to the demand regression. Institutional demand remains strongly related to lag institutional demand even after controlling for momentum. For all securities, lag institutional demand has coefficient 0.1133, while lag return has coefficient 0.0691. For return-adjusted portfolio weights, the corresponding coefficients are 0.1154 and 0.0545.

Interpretation. Institutions are momentum traders, but momentum trading explains only a small part of herding. Lag institutional demand is the stronger predictor of current institutional demand.

Table 3
Tests for herding—buyer if increased return-adjusted portfolio weight

$$\Delta_{k,t} = \beta_t \Delta_{k,t-1} + \varepsilon_{k,t}$$

Average coefficient (β)	Partitioned slope coefficient: Institutions following their own trades	Partitioned slope coefficient: Institutions following others' trades	Average R^2
Panel A: Securities with ≥ 1 institutional trader			
0.1195	0.0615	0.0580	2.38%
(9.26)**	(4.57)**	(8.61)**	
Panel B: Securities with ≥ 5 institutional traders			
0.2048	0.0675	0.1373	4.63%
(23.38)**	(9.25)**	(15.21)**	
Panel C: Securities with ≥ 10 institutional traders			
0.2476	0.0565	0.1910	6.44%
(33.67)**	(14.52)**	(26.40)**	
Panel D: Securities with ≥ 20 institutional traders			
0.2959	0.0446	0.2513	9.01%
(43.64)**	(22.23)**	(38.59)**	

For each security and quarter between March 1983 and December 1997 we calculate the fraction of institutional traders that increase their return-adjusted portfolio weight in the security. An institution is defined as increasing their return-adjusted portfolio weight if the security's weight in the investor's portfolio at the end of the quarter is larger than what it would be if the manager did not rebalance their beginning-of-quarter portfolio. All data are standardized (i.e., rescaled to zero mean, unit variance) each quarter. We then estimate 58 quarterly cross-sectional regressions of institutional demand on lag institutional demand. Because there is a single independent variable in each regression and the data are standardized, these regression coefficients are also the cross-sectional correlations between institutional demand and lag institutional demand. The first column reports the time-series average of these 58 correlation coefficients and associated *t*-statistic (in parentheses, computed from time-series standard errors). The second and third columns report the portion of the correlation [see Equation (4)] that results from institutional investors following their own lag trades and the portion that results from institutions following the previous trades of other institutions (herding). Panels B, C, and D report the averages when limiting the sample to securities with at least 5, 10, or 20 institutional traders, respectively. **Indicates statistical significance at the 1% level; *indicates statistical significance at the 5% level.

Moreover, the results are inconsistent with the hypothesis that institutional investors follow their own lag trades as a result of time-series correlation in their net flows and investing flows into their existing portfolios.[27,28]

Table 3: Tests for Herding - Buyer if Increased Return-Adjusted Portfolio Weight

Table 4
Standardized regression of institutional demand on lag institutional demand and lag return

$$\Delta_{k,t} = \beta_{1,t} \Delta_{k,t-1} + \beta_{2,t} R_{k,t-1} + \varepsilon_{k,t}$$

Average coefficient associated with lag institutional demand (β_1)	Average coefficient associated with lag return (β_2)	Average R^2
Panel A: Securities with ≥ 1 institutional trader		
Regression 1—Buyer if increased position		
0.1133	0.0691	2.59%
(11.84)**	(12.10)**	
Regression 2—Buyer if increased return-adjusted portfolio weight		
0.1154	0.0545	2.86%
(9.00)**	(9.42)**	
Panel B: Securities with ≥ 5 institutional traders		
Regression 1—Buyer if increased position		
0.1624	0.0789	4.19%
(22.25)**	(11.90)**	
Regression 2—Buyer if increased return-adjusted portfolio weight		
0.1953	0.0774	5.50%
(22.32)**	(10.89)**	
Panel C: Securities with ≥ 10 institutional traders		
Regression 1—Buyer if increased position		
0.1590	0.0671	3.90%
(26.26)**	(8.98)**	
Regression 2—Buyer if increased return-adjusted portfolio weight		
0.2340	0.0801	7.34%
(31.97)**	(11.16)**	
Panel D: Securities with ≥ 20 institutional traders		
Regression 1—Buyer if increased position		
0.1490	0.0476	3.23%
(26.90)**	(6.54)**	
Regression 2—Buyer if increased return-adjusted portfolio weight		
0.2802	0.0658	9.79%
(42.75)**	(8.04)**	

For each security and quarter between March 1983 and December 1997 we calculate the fraction of institutional traders that increase their position in the security and the fraction that increase their return-adjusted portfolio weight in the security. An investor is defined as increasing their position if they hold a greater fraction of the firm's shares at the end of the quarter than they held at the beginning. An institution is defined as increasing their return-adjusted portfolio weight if the security's weight in the investor's portfolio at the end of the quarter is larger than what it would be if the manager did not rebalance their beginning-of-quarter portfolio. We then estimate 58 quarterly cross-sectional regressions of each measure on their lag value and lag quarterly return. To allow direct comparison between the coefficients associated with the independent variables, we standardized (i.e., rescale to zero mean, unit variance) all data each quarter. The average (standardized) coefficients from the 58 cross-sectional regressions and associated *t*-statistics (computed from time-series standard errors) are reported in panel A. Panels B, C, and D report the averages when limiting the sample to securities with at least 5, 10, or 20 institutional traders, respectively. **Indicates statistical significance at the 1% level; *indicates statistical significance at the 5% level.

Table 4: Standardized Regression of Institutional Demand on Lag Institutional Demand and Lag Return

9.5 Table 5: Correlation between Institutional Demand and Returns

Read. Table 5 shows that institutional demand is positively correlated with prior-quarter and same-quarter returns. Future return correlations are weakly positive, especially for stocks with at least five, ten, or twenty institutional traders. For example, with at least ten institutional traders, institutional demand correlates 0.0291 with next-quarter returns, 0.0326 with next-six-month returns, and 0.0219 with next-year returns.

Interpretation. The absence of future return reversals argues against the view that herding is simply fads or reputational crowding that pushes prices away from fundamentals. Instead, institutional trading appears related to information entering prices.

9.6 Table 6: Herding and Following Their Own Trades by Capitalization Quintile (>5 Institutional Traders)

Read. Table 6 shows that herding and own-trade persistence exist in every capitalization quintile. In average contribution terms, the herding contribution declines monotonically from 0.0036 for the smallest firms to 0.0001 for the largest firms. The own-trade contribution also declines with firm size, from 0.0439 to 0.0255.

Interpretation. Herding is strongest in small stocks, supporting the informational cascade interpretation. Own-trade persistence is also strongest in small stocks, consistent with gradual trading to reduce market impact and liquidity costs.

Table 5
Correlation between institutional demand and returns

Sample	Previous quarter return	Same quarter return	Following quarter return	Following 6-month return	Following 12-month return
≥1 Institutional trader	0.0744	0.0784	0.0077	0.0060	− 0.0009
	(14.45)**	(16.94)**	(1.96)	(1.33)	(− 0.25)
≥5 Institutional traders	0.1116	0.1466	0.0221	0.0241	0.0139
	(17.10)**	(22.31)**	(4.36)**	(4.71)**	(3.52)**
≥10 Institutional traders	0.1101	0.1773	0.0291	0.0326	0.0219
	(14.12)**	(25.68)**	(5.17)**	(5.79)**	(4.25)**
≥20 Institutional traders	0.0998	0.2079	0.0257	0.0300	0.0205
	(12.17)**	(28.47)**	(3.47)**	(4.33)**	(3.43)**

For each security and quarter between March 1983 and December 1997 we calculate the fraction of institutional traders that increase their position in the security. An investor is defined as increasing their position if they hold a greater fraction of the firm's shares at the end of the quarter than they held at the beginning. Each quarter, we then calculate the cross-sectional correlation between the fraction of institutions buying and returns over the previous quarter, the same quarter, the following quarter, the following six months, and the following year. The time-series averages of these cross-sectional correlations and associated t-statistics (computed from time-series standard errors) are reported for the sample that includes all securities with at least one institutional trader and samples limited to securities with at least 5, 10, or 20 institutional traders. **Indicates statistical significance at the 1% level; *indicates statistical significance at the 5% level.

Table 5: Correlation between Institutional Demand and Returns

Table 6
Herding and following their own trades by capitalization quintile (≥5 institutional traders)

Panel A: Coefficient and slope decomposition

Capitalization quintile	Average coefficient (β)	Partitioned slope coefficient: Institutions following their own trades	Partitioned slope coefficient: Institutions following others' trades	Average R^2
Small firms	0.1801	0.0892	0.0909	3.65%
	(21.17)**	(15.05)**	(11.13)**	
Quintile 2	0.1469	0.0743	0.0726	2.53%
	(18.30)**	(16.23)**	(10.06)**	
Quintile 3	0.1307	0.0618	0.0690	1.95%
	(20.22)**	(17.43)**	(13.31)**	
Quintile 4	0.1368	0.0409	0.0959	2.20%
	(17.94)**	(15.36)**	(13.58)**	
Large firms	0.1644	0.0269	0.1375	3.07%
	(20.60)**	(20.17)**	(17.98)**	

Panel B: Average contributions

	Average contribution from following their own trades	Average contribution from following others' trades
Small firms	0.0439	0.0036
	(16.91)**	(10.40)**
Quintile 2	0.0372	0.0020
	(18.86)**	(9.94)**
Quintile 3	0.0340	0.0014
	(19.79)**	(12.36)**
Quintile 4	0.0285	0.0013
	(22.61)**	(12.77)**
Large firms	0.0255	0.0001
	(27.44)**	(17.15)**
F-statistic (p-value)	16.33	28.97
	(0.01)	(0.01)

For each security and quarter between March 1983 and December 1997 we calculate the fraction of institutional traders that increase their position in the security. An investor is defined as increasing their position if they hold a greater fraction of the firm's shares at the end of the quarter than they held at the beginning. Firms (with at least five institutional traders) are then sorted, each quarter, into quintiles based on beginning of quarter capitalization. Within each capitalization quintile, all data are standardized (i.e., rescaled to zero mean, unit variance) each quarter. We then estimate 58 quarterly cross-sectional regressions of the fraction of institutional traders buying on the lag fraction of institutional traders buying for firms within each capitalization quintile. Because there is a single independent variable in each regression and the data are standardized, these regression coefficients are also the cross-sectional correlations between the fraction of institutional traders buying this quarter and the fraction of institutional traders buying last quarter. The first column in panel A reports the time-series average of these 58 correlation coefficients and associated t-statistic (in parentheses, computed from time-series standard errors). The second and third columns in panel A report the portion of the correlation [see Equation (4)] that results from institutional investors following their own lag trades and the portion that results from institutions following the previous trades of other institutions (herding). We then compute the cross-sectional average contribution from following their own trades [Equation (9)] and following others' trades [Equation (10)] for firms within each capitalization quintile, each quarter. Time-series averages and associated t-statistics (computed from time-series standard errors) of these 58 cross-sectional averages are reported in panel B. The last row in panel B reports F-statistics associated with the null hypothesis that the values are equal across capitalization quintiles. **Indicates statistical significance at the 1% level; *indicates statistical significance at the 5% level.

Table 6: Herding and Following Their Own Trades by Capitalization Quintile (At Least 5 Institutional Traders)

9.7 Table 7: Analysis over Time (>5 Institutional Traders)

Read. Table 7 compares 1983-1990 with 1990-1997. For all firms with at least five institutional traders, the total coefficient declines from 0.1982 to 0.1527. The own-trade component declines from 0.0823 to 0.0524. Average contribution measures also decline over time.

Interpretation. Institutions become less likely to follow their own lag trades, consistent with lower trading costs and improved liquidity. The decline in herding is most evident in large stocks, consistent with faster information incorporation in large-capitalization securities.

9.8 Table 8: Analysis by Investor Type (>5 Institutional Traders)

Read. Table 8 decomposes persistence by institutional investor type. Every class exhibits significant own-trade persistence and herding. Bank trust departments and independent advisors show high total correlations. In average contribution terms, bank trust departments have the highest herding contribution, while mutual funds and insurance companies have weaker herding contributions.

Interpretation. Herding is not confined to one institutional group. However, differences across investor types suggest that institutional environment matters. Evidence for reputational herding is mixed because mutual funds and independent advisors, which should be particularly reputation-sensitive, do not display the strongest same-type herding.

Table 7
Analysis over time (≥5 institutional traders)

Panel A: Coefficient and slope decomposition

Sample	Average coefficient (β_t)	Partitioned slope coefficient: Institutions following their own trades	Partitioned slope coefficient: Institutions following others' trades
All firms (198309–199009)	0.1982	0.0823	0.1158
All firms (199012–199712)	0.1527	0.0524	0.1003
t-statistic	3.65**	4.57**	1.37
Small firms (198309–199009)	0.1933	0.1058	0.0857
Small firms (199012–199712)	0.1668	0.0708	0.0960
t-statistic	1.58	3.38**	− 0.63
Quintile 2 (198309–199009)	0.1683	0.0932	0.0751
Quintile 2 (199012–199712)	0.1255	0.0553	0.0702
t-statistic	2.83**	4.91**	0.34
Quintile 3 (198309–199009)	0.1456	0.0777	0.0679
Quintile 3 (199012–199712)	0.1159	0.0458	0.0700
t-statistic	2.39*	5.55**	− 0.20
Quintile 4 (198309–199009)	0.1682	0.0508	0.1175
Quintile 4 (199012–199712)	0.1055	0.0301	0.0744
t-statistic	4.86**	4.20**	3.30**
Large firms (198309–199009)	0.1707	0.0269	0.1438
Large firms (199012–199712)	0.1581	0.0270	0.1312
t-statistic	0.79	− 0.02	0.82

Panel B: Average contributions

	Average contribution from following their own trades	Average contribution from following others' trades
All firms (198309–199009)	0.0401	0.0028
All firms (199012–199712)	0.0284	0.0021
t-statistic	4.60**	2.26**
Small firms (198309–199009)	0.0523	0.0035
Small firms (199012–199712)	0.0355	0.0037
t-statistic	3.57**	− 0.30
Quintile 2 (198309–199009)	0.0456	0.0022
Quintile 2 (199012–199712)	0.0287	0.0018
t-statistic	5.16**	0.90
Quintile 3 (198309–199009)	0.0421	0.0015
Quintile 3 (199012–199712)	0.0259	0.0013
t-statistic	5.95**	0.80
Quintile 4 (198309–199009)	0.0321	0.0017
Quintile 4 (199012–199712)	0.0249	0.0010
t-statistic	3.08**	4.12**
Large firms (198309–199009)	0.0242	0.0011
Large firms (199012–199712)	0.0269	0.0009
t-statistic	− 1.43	2.31*

For each security and quarter between March 1983 and December 1997 we calculate the fraction of institutional traders that increase their position in the security. Firms (with at least five institutional traders) are then sorted, each quarter, into quintiles based on beginning of quarter capitalization. Within each capitalization quintile, all data are standardized (i.e., rescaled to zero mean, unit variance) each quarter. We then estimate 58 quarterly cross-sectional regressions of the fraction of institutional traders buying on the lag fraction of institutional traders buying for firms within each capitalization quintile. We also repeat the analysis for the sample including all firms (with at least five institutional traders). Because there is a single independent variable in each regression and the data are standardized, these regression coefficients are also the cross-sectional correlations between the fraction of institutional traders buying this quarter and the fraction of institutional traders buying last quarter. Panel A reports the time-series average of these correlation coefficients and the portion of the correlation [see Equation (4)] that results from institutional investors following their own lag trades and the portion that results from institutions following the previous trades of other institutions (herding) over the first 29 quarters in the sample period (198309–199009) and the second 29 quarters in the sample period (199012–199712). The third row reports t-statistics from a difference in means test of the hypothesis that the mean value in the first period equals the mean value in the second period. Panel B repeats the analysis for the average contribution from following their own trades [Equation (9)] and following others' trades [equation (10)] for firms within each capitalization quintile and the sample including all firms (with at least five institutional traders). **Indicates statistical significance at the 1% level; *indicates statistical significance at the 5% level.

Table 7: Analysis over Time (At Least 5 Institutional Traders)

Table 8
Analysis by investor type (≥5 institutional traders)

Panel A: Coefficient and slope decomposition

		Partitioned slope coefficient				
			Institutions following others' trades			
Trader type	Average coefficient (β)	Following their own trades	All	Same type	Different type	Average R^2
Banks	0.1293	0.0207	0.1086	0.0499	0.0587	1.92%
	(19.55)**	(8.15)**	(18.94)**	(15.25)**	(15.61)**	
Insurance companies	0.0495	0.0164	0.0331	0.0047	0.0284	0.45%
	(8.29)**	(5.75)**	(4.84)**	(4.23)**	(4.42)**	
Mutual funds	0.0521	0.0134	0.0387	0.0029	0.0358	0.57%
	(7.14)**	(8.69)**	(5.56)**	(2.35)*	(5.54)**	
Independent advisors	0.1338	0.0587	0.0750	0.0344	0.0407	1.98%
	(23.01)**	(26.03)**	(13.24)**	(11.55)**	(10.86)**	
Unclassified	0.0469	0.0081	0.0388	0.0068	0.0320	0.66%
	(5.32)**	(4.85)**	(4.36)**	(3.42)**	(3.90)**	

Panel B: Average contributions

	Average contribution from following their own trades	Average contribution from following others' trades	Average contribution from following same type traders	Average contribution from following different type traders	Average "same contribution" less average "different contribution"
Banks	0.0200	0.0023	0.0033	0.0018	0.0015
	(11.97)**	(18.44)**	(14.06)**	(14.37)**	(6.66)**
Insurance companies	0.0248	0.0006	0.0012	0.0006	0.0006
	(13.18)**	(5.99)**	(5.46)**	(5.33)**	(3.06)**
Mutual funds	0.0347	0.0006	0.0009	0.0006	0.0002
	(22.05)**	(5.11)**	(2.83)**	(4.98)**	(0.81)
Independent advisors	0.0367	0.0016	0.0016	0.0017	−0.0001
	(28.13)**	(12.51)**	(11.40)**	(10.43)**	(−0.38)
Unclassified	0.0410	0.0017	0.0015	0.0019	−0.0005
	(25.10)**	(8.99)**	(7.48)**	(8.57)**	(−2.35)*
F-statistic	25.71	34.96	17.49	22.05	
(p-value)	(0.01)	(0.01)	(0.01)	(0.01)	

For each security and quarter between March 1983 and December 1997 we calculate the fraction of institutional traders overall and by each investor type (bank trust departments, insurance companies, mutual funds, independent investment advisors, and unclassified institutional investors) that increase their position in the security. We then restrict the sample to securities with at least five bank trust departments trading this quarter and five institutional investors (of any classification) trading last quarter and estimate 58 quarterly cross-sectional regressions of the fraction of bank trust department traders buying on the lag fraction of all institutional traders buying. Because there is a single independent variable in each regression and the data are standardized, these regression coefficients are also the cross-sectional correlations between the fraction of bank trust department traders buying this quarter and the fraction of all institutional traders buying last quarter. The first column in panel A reports the time-series average of these 58 correlation coefficients and associated t-statistic (in parentheses, computed from time-series standard errors). The second and third columns report the portion of the correlation [see Equation (4)] that results from bank trust departments following their own lag trades and the portion that results from bank trust departments following the previous trades of other institutions (herding). The fourth and fifth columns report the portion of the coefficient that results from bank trust departments following other bank trust department trades and the portion that results from bank trust departments following non-bank trust department trades, respectively [see Equation (12)]. The analysis is then repeated for each of the other four institutional classifications. Panel B repeats the analysis for the average contribution from following their own trades [Equation (9)], following others' trades [Equation (10)], following similarly classified traders [Equation (13)], and following differently classified traders [Equation (14)]. The last column in Panel B reports the average difference between columns three and four (and associated t-statistic computed from a paired t-test). The last row in panel B reports F-statistics associated with the null hypothesis that values are equal across investor classifications. **Indicates statistical significance at the 1% level; *indicates statistical significance at the 5% level.

Table 8: Analysis by Investor Type (At Least 5 Institutional Traders)

10 Key takeaways

1. Institutional investors herd: they follow other institutions into and out of the same securities over adjacent quarters.

2. Institutional demand persistence is about half own-trade continuation and about half following other institutions' trades.
3. Herding is not mainly a mechanical result of fund flows, because it remains when using return-adjusted portfolio weights.
4. Herding is not mainly momentum trading, because lag institutional demand predicts current institutional demand more strongly than lag returns.
5. The lack of future return reversals suggests that herding is not simply destabilizing fad-based trading.
6. Herding is strongest in small stocks, consistent with informational cascades in settings where individual private signals are noisier.
7. Institutions follow their own lag trades especially in smaller stocks, consistent with gradual trading to reduce price impact and liquidity costs.
8. Herding varies by institutional type, with bank trust departments showing especially strong herding contributions and mutual funds showing relatively weak herding contributions.

11 How to use this paper

This paper is useful for three broad literatures. For asset pricing, it provides evidence that institutional trades are persistent and potentially information-relevant. For market microstructure, it shows that institutions appear to split trading programs across quarters, especially in smaller stocks. For delegated portfolio management, it provides evidence that institutional behavior is correlated across managers, but not in a way that is easily reduced to reputation or fads.

The most important methodological lesson is that herding should be measured dynamically. A within-period imbalance measure can miss interquarter following behavior. Directly estimating whether demand follows lag demand, and decomposing that persistence into own-trade and other-trade components, gives a sharper view of institutional trading dynamics.